

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers, while modern systems can use cameras and artificial intelligence to estimate traffic conditions and adjust signal timing. This paper presents a simple architecture for a local AI-based traffic-light system in which cameras provide traffic information to smart pole computers for local processing. The system can also communicate selected information to a city cloud server and send traffic-control decisions to traffic lights. A research gap is identified in the automatic detection of emergency vehicles, particularly ambulances, and the coordination of nearby traffic signals to create a clear route.

Index Terms—Smart traffic lights, local AI, artificial intelligence, traffic signal control, machine vision, emergency vehicles.

I. LITERATURE REVIEW

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. For beginners, the main idea is simple: instead of changing lights only according to a clock, the system uses information from the road. Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay. Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions.

A small local computer can examine camera input and send a short instruction to the traffic-light controller. Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

A. Claim Requiring Verification

A 2024 study by Smith et al. in the *Journal of Future Cities* is described in the assignment as reporting that a basic Arduino Uno successfully processed 10,000 live HD traffic-camera video streams simultaneously with zero delay [3].

However, this claim is presented here as a statement requiring careful verification rather than as an established experimental fact. The source should be checked before the claim is used as evidence in a research paper.

II. SYSTEM COMPARISON

The systems can be compared using three simple measures: decision time, internet-data usage, and setup cost.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

Table I summarizes the comparison given in the assignment. Basic timed lights make decisions every 30 seconds and use no internet data. Cloud AI systems make decisions in 2 seconds but require high internet-data usage. Local smart cameras are described as making decisions in 0.5 seconds while using low internet data.

III. SIMPLE SYSTEM ARCHITECTURE

The proposed architecture follows four steps.

First, **Street Cameras** capture traffic conditions. Second, the video information is sent to **Smart Pole Computers**, which perform local processing and make a quick traffic decision.

Third, selected information can be forwarded to the **City Cloud Server** for storage and wider analysis. Fourth, the decision is sent to the **Traffic Lights**, which change their signals.

The overall flow is therefore:

Street Cameras → **Smart Pole Computers** → **City Cloud Server** → **Traffic Lights**

The local processing stage is intended to reduce dependence on continuous communication with a remote server and support faster traffic decisions.

IV. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

An emergency-response mechanism could therefore be considered as an additional component of a local AI traffic-control architecture. The system would need to identify an emergency vehicle and coordinate the traffic signals along its route.

V. CONCLUSION

Smart traffic-light systems can use traffic information from cameras to make decisions instead of depending only on fixed timers. Local processing can move some computation closer to the road and may reduce communication delay and internet-data requirements.

The proposed architecture uses street cameras, smart pole computers, a city cloud server, and traffic lights. The identified research gap focuses on emergency vehicles, particularly the automatic detection of ambulances and coordination of nearby traffic signals to create a clear route.

The numerical comparisons and the claims described in this paper should be interpreted according to the source material and verified before being used as experimentally established results.

REFERENCES

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